

Test 8 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.DP.1.5 moderate	2, 4, 2, 4	Count how many values fall in each interval, then set that bar height.
2	MA.6.NSO.4.2 moderate	-16	A loss each week is a negative change repeated, so multiply.
3	MA.6.DP.1.3 moderate	25	Read the value at that mark on the box plot.
4	MA.6.GR.1.1 easy	(2, -2)	Reflecting across the y-axis changes the sign of the x-coordinate only.
5	MA.6.NSO.1.4 challenging	-47	Farthest from sea level means the greatest absolute value.
6	MA.6.AR.3.3 moderate	28	Both quantities are multiplied by 4.
7	MA.6.NSO.3.4 moderate	$2^2 \times 3 \times 7$	Divide by the smallest prime repeatedly until only primes remain.
8	MA.6.AR.2.1 easy	15	Only numbers greater than 14 make the inequality true, and 14 itself does not.
9	MA.6.NSO.3.2 moderate	18	Find the largest number that divides both addends.
10	MA.6.NSO.2.2 challenging	$\frac{9}{40}$	Multiply numerator by numerator and denominator by denominator, then simplify.
11	MA.6.NSO.4.1 moderate	0	Add the rise and subtract the fall from the starting temperature.
12	MA.6.AR.3.1 moderate	8:5	Divide both numbers by their greatest common factor.
13	MA.6.NSO.1.3 challenging	8.7	Absolute value is the distance from 0, so it is never negative.
14	MA.6.NSO.2.3 challenging	\$17.17	Multiply the price by the number of items, then add the one-time fee.
15	MA.6.AR.1.4 easy	$15x + 12$	Only like terms can be combined, so add the x-terms and leave the constant.
16	MA.6.AR.2.3 easy	$5x = 30$	The number of baskets times the guavas in each basket is the total.
17	MA.6.NSO.3.5 moderate	75	Multiply the decimal by 100 to write it as a percent.
18	MA.6.NSO.3.1 moderate	6	The greatest number of identical bags is the greatest common factor.
19	MA.6.AR.3.4 moderate	22.5	Find the discount, then subtract it from the original price.

#	Benchmark	Answer	Explanation
20	MA.6.DP.1.1 easy	How old is my teacher?	A question with only one possible answer does not produce a data set.
21	MA.6.AR.1.2 easy	$x \geq 15$	A closed circle includes the endpoint; an open circle does not. The shading shows which direction makes the inequality true.
22	MA.6.GR.2.3 easy	6	Divide the volume by the area of the base to find the height.
23	MA.6.NSO.1.1 challenging	-2.5	On a number line the least number is the one farthest to the left.
24	MA.6.AR.3.5 moderate	18	Each yard is 3 feet, so multiply by 3.
25	MA.6.DP.1.6 moderate	23 $\frac{1}{2}$	Reorder the remaining values and find the middle; with six values, average the two in the middle.
26	MA.6.NSO.3.1 challenging	24	List the factors of each number and choose the largest one they share.
27	MA.6.NSO.3.3 moderate	0.09	Squaring means multiplying the number by itself.
28	MA.6.NSO.1.2 challenging	-7	Below the surface is negative: $-19 + 12 = -7$.
29	MA.6.NSO.2.1 challenging	8.525	Multiply as with whole numbers, then place the decimal point using the total number of decimal places in the factors.
30	MA.6.GR.2.4 easy	240	Add the square base to the four triangular faces, each with area one half times base times slant height.
31	MA.6.GR.2.1 easy	10	Work backwards: double the area, then divide by the height.
32	MA.6.AR.2.2 easy	$x - 3 = 26$	Starting amount minus the stickers given away equals what is left.
33	MA.6.AR.3.2 moderate	4.76	Divide the total cost by the number of bottles.
34	MA.6.AR.1.3 easy	12	Work inside the parentheses first, then divide.
35	MA.6.GR.1.3 easy	15	Find the side lengths by subtracting coordinates, then apply the formula.
36	MA.6.AR.1.1 easy	the quotient of a number x and 6	Read the expression one operation at a time and match it to the words.
37	MA.6.GR.2.2 easy	100	Decompose the trapezoid into a rectangle and triangles, or average the two bases and multiply by the height.
38	MA.6.DP.1.4 moderate	The data are skewed to the right, with most values in the lower intervals.	Most of the data sit in the low intervals with a tail stretching to the right, so the distribution is skewed right.